

Корни из единицы и комплексная экспонента

1 сентября 2026 года

Теоретический минимум.

Опр. Let $\varepsilon = e^{2\pi i/n}$. The numbers $1, \varepsilon, \dots, \varepsilon^{n-1}$ are the roots of $z^n - 1$, so $\sum_{k=0}^{n-1} \varepsilon^{km}$ equals n if $n \mid m$ and 0 otherwise (*фильтр по корням из единицы*), and $z^n - 1 = \prod_{k=0}^{n-1} (z - \varepsilon^k)$.

Теорема 1 (произведение диагоналей) $\prod_{k=1}^{n-1} |1 - \varepsilon^k| = \prod_{k=1}^{n-1} 2 \sin \frac{k\pi}{n} = n$: the product of the distances from one vertex of a regular n -gon inscribed in the unit circle to all the others equals n .

Теорема 2 (вещественная часть кругового поля) $\mathbb{Q}(\cos \frac{2\pi}{n})$ is the real subfield of $\mathbb{Q}(\varepsilon)$; it has degree $\varphi(n)/2$ over $\mathbb{Q}$ for $n > 2$, and $2 \cos \frac{2\pi}{n} = \varepsilon + \varepsilon^{-1}$ is an algebraic integer.

Факт 1 Every symmetric expression in ε^k and ε^{-k} is a polynomial in $2 \cos \frac{2\pi k}{n}$ with integer coefficients: this is the bridge between «тригонометрия» and «теория чисел».

1. Prove the root-of-unity filter and use it to count the subsets of $\{1, \dots, n\}$ whose sum is divisible by 3; then evaluate $\sum_{k \equiv 0 \pmod 3} \binom{n}{k}$ in closed form.
2. Prove Theorem 1, and deduce $\sum_{k=1}^{n-1} \frac{1}{|1 - \varepsilon^k|^2} = \frac{n^2 - 1}{12}$.
3. (CIIM 2009) Let ε be an n -th root of unity and suppose that $z = p(\varepsilon)$ is a real number for some $p \in \mathbb{Z}[x]$. Prove that there is $q \in \mathbb{Z}[x]$ with $z = q(2 \cos \frac{2\pi}{n})$.
4. (Мехмат, 2009) Let n be odd. On the sides of an arbitrary n -gon isosceles triangles with apex angle $\frac{2\pi}{n}$ are erected outwards; their apices form a second n -gon. On its sides one erects isosceles triangles with

apex angle $\frac{4\pi}{n}$, and so on: the $(k+1)$ -st n -gon is obtained from the k -th one with the angle $\frac{2\pi k}{n}$, the triangles being erected outwards when $\frac{2\pi k}{n} < \pi$ and inwards when $\frac{2\pi k}{n} > \pi$. Prove that the $(n-1)$ -st n -gon is regular. (*Подсказка: постройка — это умножение вершинного вектора на $1+i\cot\frac{\pi k}{n}$, разложенное по базису корней из единицы.*)

5. (Proposed problems, №232) Let $A \in M_n(\mathbb{C})$ satisfy $A^m = (-1)^m E$ for some $m \geq 2$, and put $\varepsilon = \cos \frac{2\pi}{m} + i \sin \frac{2\pi}{m}$. Prove that

$$n \leq \text{rank}(E+A) + \text{rank}(E+\varepsilon A) + \cdots + \text{rank}(E+\varepsilon^{m-1}A) \leq (m-1)n.$$